

A Paley–II analogue of the Scarpis–Đoković construction for prime powers $q \equiv 1 \pmod{4}$

Jeffery Kline

July 18, 2026

Abstract

A Hadamard matrix of order m is a $\{1, -1\}$ -matrix H with $HH^T = mI_m$. Scarpis (1898) showed that for a prime $p \equiv 3 \pmod{4}$ a Hadamard matrix of order $p + 1$ can be lifted to one of order $p(p + 1)$, and Đoković extended the lift to all prime powers $q \equiv 3 \pmod{4}$, asking for the analogous procedure for $q \equiv 1 \pmod{4}$. That case was settled by Farouk and Wang (2020), who lift an input Hadamard matrix of order $2(q + 1)$, subject to eligibility conditions, to one of order $2q(q + 1)$, verifying orthogonality by row-pair counting arguments in which the key inner products are imported from the Hadamard property of the input matrix. This note contributes a different method, not a new existence result: there is no input matrix; the finite bands are given entrywise in closed form by the quadratic character on $\mathbb{F}_q$ composed with the 2×2 Paley–II replacement blocks, and the Hadamard property is proved by four block-Gram identities computed directly from character sums. The outputs differ as well: at $q = 5$ our matrix is not Hadamard-equivalent to the order-60 example printed by Farouk and Wang. We also make the family’s coverage explicit: every order $2q(q + 1)$ whose odd part is at most 3000 is already recorded in the standard tables, and the smallest order lying beyond that audited range, $q = 81$ (order $13284 = 4 \cdot 41 \cdot 81$), also follows from a classical Turyn product using T -matrices of order 41 and Williamson-type matrices of order $81 = 9^2$. We exhibit the witnessing constructions, identify the smallest unaudited order, and tabulate those orders up to $q = 2917$ for which no elementary witness is presently evident, emphasising that these are *unaudited* rather than open.

1 Introduction

For a prime power q let $\mathbb{F} = \mathbb{F}_q$ and let $\chi: \mathbb{F} \rightarrow \{0, \pm 1\}$ be the quadratic character, $\chi(0) = 0$, $\chi(x) = 1$ for nonzero squares and -1 otherwise. Paley’s two constructions split on the residue of q : for $q \equiv 3 \pmod{4}$ Paley I gives a Hadamard matrix of order $q + 1$, and for $q \equiv 1 \pmod{4}$ Paley II gives one of order $2(q + 1)$ [4].

Scarpis [5] showed that for a prime $p \equiv 3 \pmod{4}$ a Hadamard matrix of order $p + 1$ lifts to one of order $p(p + 1)$. Đoković [9] replaced the prime by an arbitrary prime power: if $q \equiv 3 \pmod{4}$ is a prime power there is a Hadamard matrix of order $q(q + 1)$; his note closes with the open problem of an analogous procedure for $q \equiv 1 \pmod{4}$.

Because the Paley–II seed has order $2(q + 1)$, the natural Scarpis-type target in the $q \equiv 1 \pmod{4}$ case is

$$q \cdot 2(q + 1) = 2q(q + 1).$$

We construct such a matrix directly from χ . Throughout, $q \equiv 1 \pmod{4}$ is a prime power, so $\chi(-1) = 1$; this is the only place the residue class is used, and it is exactly what fails for $q \equiv 3 \pmod{4}$.

The existence of Hadamard matrices of order $2q(q+1)$ for prime powers $q \equiv 1 \pmod{4}$ is due to Farouk and Wang [1]; the contribution of this note is the method of construction and proof. Farouk and Wang obtain the matrix as the output of a lifting procedure applied to an input Hadamard matrix of order $2(q+1)$ meeting certain eligibility conditions, and their verification proceeds row pair by row pair, by counting arguments in which the key inner-product values are imported from the Hadamard property of the input matrix. Here there is no input matrix: the finite bands are written entrywise in the closed form $\psi(\chi(y-x-t))$, and the Hadamard property follows from four block-Gram identities whose ingredients are computed directly from the quadratic character — the sums (2.1) and $\sum_z \chi(z(z-d)) = -1$, together with the block identity $P + ZR = 0$. The proof is thus self-contained at the level of character sums, and the single place where the residue class enters, $\chi(-1) = 1$, is isolated. A secondary contribution is the coverage analysis of Appendix A, which has no counterpart in [1]: it locates every order of the family against the audited tables of known Hadamard orders.

2 The Paley–II blocks

Put

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and let $\psi: \{0, \pm 1\} \rightarrow \{\pm 1\}^{2 \times 2}$ be $\psi(0) = Z$, $\psi(1) = P$, $\psi(-1) = -P$; these are the Paley–II replacement blocks, and $ZR = -P$. Write $\mathcal{R} = J_q \otimes I_2$, where J_q is the $q \times q$ all-ones matrix. We use repeatedly

$$\sum_{u \in \mathbb{F}} \psi(\chi(u)) = Z, \tag{2.1}$$

which holds because the nonzero squares and non-squares are equinumerous.

3 The construction

For $t \in \mathbb{F}$ let K_t be the $2q \times 2q$ matrix with rows and columns indexed by pairs (x, a) , $x \in \mathbb{F}$, $a \in \{0, 1\}$, whose 2×2 block in position (x, y) is

$$K_t[x, y] = \psi(\chi(y - x - t)).$$

Lemma 1. *For every $t \in \mathbb{F}$, $K_t K_t^T = 2(q+1)I_{2q} - 2\mathcal{R}$.*

Proof. The (x, x') block of $K_t K_t^T$, with $d = x' - x$ and $z = y - x - t$, equals $\sum_z \psi(\chi(z))\psi(\chi(z-d))^T$. For $d = 0$ each summand is $2I_2$, giving $2qI_2$. For $d \neq 0$ the terms $z = 0$ and $z = d$ carry opposite skew parts and cancel, using $\chi(-d) = \chi(d)$ (here $\chi(-1) = 1$); for $z \neq 0, d$, $\psi(\chi(z))\psi(\chi(z-d))^T = 2\chi(z)\chi(z-d)I_2$, and $\sum_z \chi(z(z-d)) = -1$ yields $-2I_2$. These are the diagonal and off-diagonal blocks of $2(q+1)I - 2\mathcal{R}$. $\square$

Lemma 2. *For $r, s \in \mathbb{F}$,*

$$\sum_{a \in \mathbb{F}} K_{ar} K_{as}^T = \begin{cases} 2q(q+1)I_{2q} - 2q\mathcal{R}, & r = s, \\ 2\mathcal{R}, & r \neq s. \end{cases}$$

Proof. The case $r = s$ is q copies of Lemma 1. For $r \neq s$, in the (x, x') block put $z = y - x - ar$; the second argument becomes $z - (x' - x) + a(r - s)$, which ranges over all of $\mathbb{F}$ independently of z as a does. By (2.1) each block is $ZZ^T = 2I_2$, i.e. the corresponding block of $2\mathcal{R}$. $\square$

For $r \in \mathbb{F}$ let B_r be the $2q \times 2q$ matrix whose row (x, a) equals row (r, a) of K_0 for every x ; thus B_r has only two distinct rows. From Lemma 1, $B_r B_s^T = 2q\mathcal{R}$ if $r = s$ and $-2\mathcal{R}$ if $r \neq s$.

Form the $2q \times 2q(q+1)$ block rows $M_r = [B_r \mid K_{ar} \ (a \in \mathbb{F})]$.

Lemma 3. $M_r M_s^T = 2q(q+1)I_{2q}$ if $r = s$, and 0 if $r \neq s$.

Proof. $M_r M_s^T = B_r B_s^T + \sum_a K_{ar} K_{as}^T$. For $r = s$ the $2q\mathcal{R}$ of the border cancels the $-2q\mathcal{R}$ of Lemma 2; for $r \neq s$ the $-2\mathcal{R}$ cancels the $2\mathcal{R}$. $\square$

Let C be the Paley conference matrix indexed by $\{\infty\} \cup \mathbb{F}$, namely $C_{u,u} = 0$, $C_{\infty,x} = C_{x,\infty} = 1$, $C_{x,y} = \chi(y-x)$, and let S be the order- $2(q+1)$ Paley-II Hadamard matrix obtained by replacing $0, 1, -1$ by $Z, P, -P$. Delete the two rows of S indexed by ∞ ; in each remaining row repeat the ∞ -column pair q times unchanged, and replace each finite column pair by its product with R , repeated q times. This is a $2q \times 2q(q+1)$ matrix M_∞ .

Lemma 4. $M_\infty M_\infty^T = 2q(q+1)I_{2q}$ and $M_\infty M_r^T = 0$ for all $r \in \mathbb{F}$.

Proof. The first identity holds because S is Hadamard, each column pair is repeated q times, and right multiplication by R preserves row inner products. For the cross identity, summing a cap row against a row of M_r and using (2.1) on every $2q$ -block leaves $(P_\alpha + (\sum_a S[e, a]_\alpha)R) \cdot Z_\beta$; the inner sum of finite entries of a row of S is Z_α , and $P + ZR = P - P = 0$. $\square$

Theorem 1. *Let $q \equiv 1 \pmod{4}$ be a prime power. Then*

$$H = \begin{bmatrix} M_\infty \\ M_r \ (r \in \mathbb{F}) \end{bmatrix}$$

is a Hadamard matrix of order $2q(q+1)$.

Proof. H is $\{1, -1\}$ and square of order $2q + q \cdot 2q = 2q(q+1)$. Lemmas 3 and 4 give $HH^T = 2q(q+1)I$. $\square$

Corollary 1. *For every prime power $q \equiv 1 \pmod{4}$ there is a Hadamard matrix of order $2q(q+1)$; the smallest non-prime instance is $q = 9$, of order 180.*

The construction has been verified by full Gram computation for all $q \leq 49$ (orders $60, \dots, 4900$, including the extension fields $q = 9, 25, 49$) and by streamed row-orthogonality checks for $q = 81$ and $q = 125$ (orders 13284 and 31500).

4 Relation to the Farouk–Wang construction

Theorem 1 gives, for the $q \equiv 1 \pmod{4}$ case, an explicit form of the Scarpis-type family that Đoković [9] asked for. That family was already constructed by Farouk and Wang [1, 2] from the same order- $2(q+1)$ Paley-II seed. This section records how the two routes differ and where the orders of the family sit relative to the audited tables of known Hadamard orders.

While Theorem 1 and [1, Cor. 2.1] assert the same existence result, the routes differ. Farouk and Wang obtain the order- $2q(q+1)$ matrix from a lifting procedure applied to a Hadamard matrix of order $2(q+1)$ satisfying the eligibility conditions of [1, Lem. 2.2] — a formally larger input class, with Paley II the only exhibited member — and verify orthogonality row pair by row pair through counting arguments (row sums, balanced sign-pair tallies, and a unique-intersection computation), importing the key inner-product values from the Hadamard property of the input matrix. The

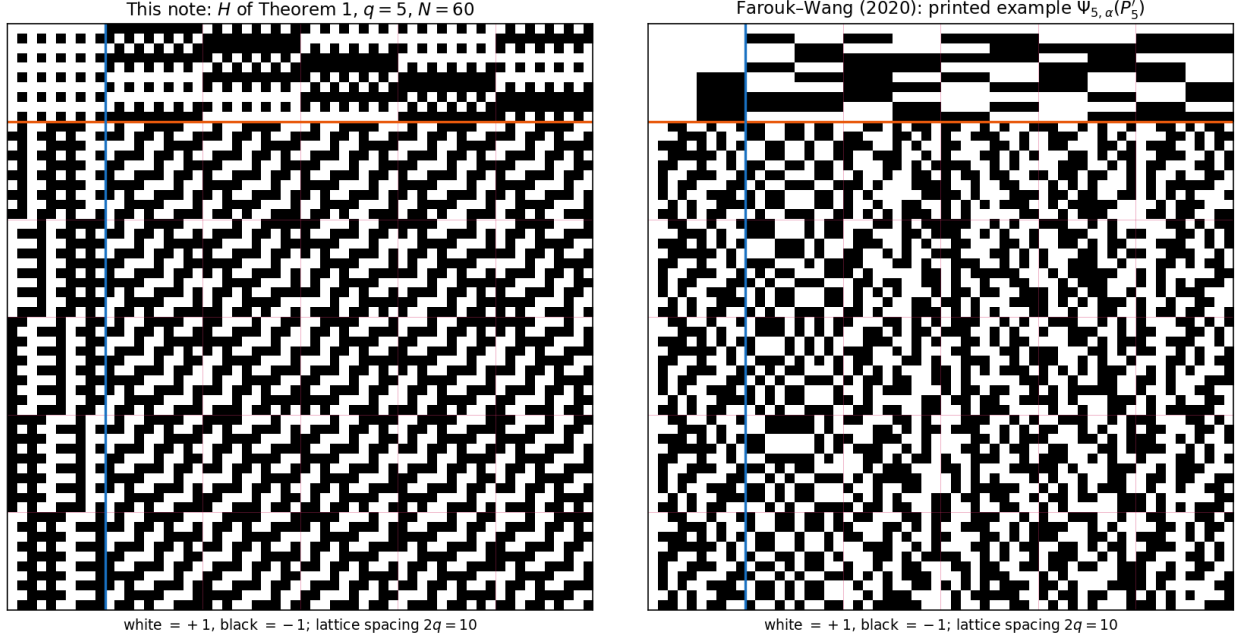


Figure 1: The two constructions at $q = 5$ (order 60; white = +1, black = -1; lattice spacing $2q = 10$). Left: the matrix of Theorem 1 — the cap band M_∞ above the orange separator is a family of period-two column patterns, and the finite bands M_r carry the diagonal texture of the shifts $\psi(\chi(y - x - ar))$. Right: the order-60 example $\Psi_{5,\alpha}(P'_5)$ printed in the appendix of [1], whose top band is a coarse tiling (its border block consists of four constant 5×5 quadrants) and whose finite bands show no diagonal structure. The two matrices are Hadamard-inequivalent: over the $\binom{60}{4} = 487,635$ row quadruples, the right matrix attains $|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}| = 28$ on 400 quadruples, the left on none, and this multiset is invariant under row and column permutations and negations (and the comparison fails in both orientations, covering the transpose convention as well). This separates the printed representative only: outputs of the Farouk–Wang procedure for other choices of α are not addressed.

present note fixes the Paley–II seed and writes the finite bands entrywise in the closed form $\psi(\chi(y - x - t))$; the Hadamard property follows from four block-Gram identities in which the corresponding quantities are computed directly from the character sums (2.1) and $\sum_z \chi(z(z - d)) = -1$, together with $P + ZR = 0$ and the Hadamard property of the Paley–II seed S (itself a character-sum identity). The correspondence is exact: their unique-intersection index matches the affine collision $x + ar = x' + as$ in the substitution step of Lemma 2, and their imported inner products match the values that Lemma 1 derives from $\sum_z \chi(z(z - d)) = -1$. The devices used for the cap band also differ: a column permutation arranging balanced sign pairs in [1] versus right multiplication by R here. The two constructions moreover produce genuinely different matrices, not merely different presentations: at $q = 5$ the matrix of Theorem 1 is not Hadamard-equivalent to the order-60 example printed in the appendix of [1] — the multiset of absolute row-quadruple correlations $|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}|$, an invariant of Hadamard equivalence, differs for the two matrices (Figure 1). Since Farouk and Wang note that their output may vary with the choice of bijection α , we leave open whether the matrix of Theorem 1 is equivalent to some other output of their procedure.

Since $q \equiv 1 \pmod{4}$, the 2-adic valuation of the order is exactly 2:

$$2q(q + 1) = 4 \cdot q \cdot \frac{q+1}{2}, \quad q \text{ and } \frac{q+1}{2} \text{ both odd.} \quad (4.1)$$

Thus every order is $4 \times (\text{odd})$, so it is never a Kronecker product of two smaller Hadamard matrices, and its odd part $q(q+1)/2$ is always composite. Two consequences follow. First, the family never meets the dominant list of open Hadamard orders, which are $4 \times \text{prime}$. Second, (4.1) has the form needed for Turyn's construction: when T -matrices of order t and Williamson-type matrices of order w are available, one obtains a Hadamard matrix of order $4tw$ [7, 8], and here $\{t, w\} = \{q, (q+1)/2\}$. The coverage analysis of Appendix A makes both points quantitative: all orders with odd part ≤ 3000 are recorded in the standard tables, the first order beyond that range is covered by such a Turyn product, and the orders for which no elementary witness is currently evident are listed but are unaudited rather than open.

A Coverage analysis

Write $N = 2q(q+1)$ and $m = q(q+1)/2$ for its odd part, so $N = 4m$. We assess coverage of N by prior constructions on three levels: the audited tables (§A.1), the first unaudited order (§A.2), and an elementary-witness test that isolates candidate orders (§A.3).

A.1 Orders with odd part at most 3000

The surveys of Seberry and Yamada [8, 11] and the database of Cati and Pasechnik [12] record, for every odd $m \leq 3000$, the least t for which a Hadamard matrix of order $2^t m$ is known. The default is $t = 2$ (so $N = 4m$ is known), and the only odd $m \leq 3000$ with $t > 2$ are the primes 167, 179, 223, 283 (orders 668, 716, 892, 1132, still open). Since $m = q(q+1)/2$ is composite it is never one of these; hence *every N with $m \leq 3000$ is a known Hadamard order*. This is the range $q \leq 73$, tabulated below with an explicit elementary witness where one exists (Paley I/II, or a Turyn product $4tw$).

q	$N = 2q(q+1)$	odd part m	witnessing construction
5	60	$3 \cdot 5$	Paley II, $p = 29$
9	180	$3^2 \cdot 5$	Paley II, $p = 89$
13	364	$7 \cdot 13$	Paley II, $p = 181$
17	612	$3^2 \cdot 17$	Turyn $4 \cdot 9 \cdot 17$
25	1300	$5^2 \cdot 13$	Turyn $4 \cdot 13 \cdot 25$
29	1740	$3 \cdot 5 \cdot 29$	Turyn $4 \cdot 15 \cdot 29$
37	2812	$19 \cdot 37$	Turyn $4 \cdot 19 \cdot 37$
41	3444	$3 \cdot 7 \cdot 41$	Paley II, $p = 1721$
49	4900	$5^2 \cdot 7^2$	Turyn $4 \cdot 25 \cdot 49$
53	5724	$3^3 \cdot 53$	Paley II, $p = 2861$
61	7564	$31 \cdot 61$	Turyn $4 \cdot 31 \cdot 61$
73	10804	$37 \cdot 73$	tables; no elementary witness (see §A.3)

The entry $q = 73$ is instructive: its odd part $2701 = 37 \cdot 73$ carries the same obstruction we use below (the prime 73), yet $N = 10804$ is a known Hadamard order because $m \leq 3000$ is audited. Obstruction in the elementary test is therefore a property of that test, not of Hadamard existence.

A.2 The first unaudited order: $q = 81$

The smallest order of the family whose odd part exceeds 3000, and hence lies beyond the audited range, is

$$q = 81 = 3^4, \quad N = 13284, \quad m = 3321 = 3^4 \cdot 41 > 3000.$$

It is nevertheless a known Hadamard order. By (4.1),

$$13284 = 4 \cdot 41 \cdot 81 = 4tw, \quad t = 41, \quad w = 81,$$

and both ingredients are classical: T -matrices of order 41 come from the base sequences $BS(21, 20)$ [7, 10], and Williamson-type matrices of order $81 = 9^2$ come from Turyn's 9-power family [7, 8]. The Turyn array on these [8, 11] yields a Hadamard matrix of order $4 \cdot 41 \cdot 81 = 13284$. Thus $q = 81$ is the smallest unaudited order, and it is covered.

A.3 An elementary-witness test and the candidate orders

Beyond the audited range we have no table to appeal to, so we apply a deliberately conservative *elementary-witness* test based on (4.1) and Turyn's array. Call an odd prime p *T-eligible* if $p \leq 59$ (so that T -matrices of order p follow from base sequences $BS(m+1, m)$, $m \leq 29$) and *W-eligible* if $p \leq 33$ or $2p - 1$ is a prime power (Williamson-type matrices of order p , the latter by Turyn's $(P+1)/2$ family with $P = 2p - 1 \equiv 1 \pmod{4}$); both classes are multiplicatively closed. An order $N = 4m$ passes the test, and is then a Hadamard order, if N is a Paley I/II order or if every prime factor of m is *T*- or *W*-eligible. A prime factor p of m is *obstructing* if $p > 59$ and $2p - 1$ is not a prime power.

A passing verdict is a genuine construction; a failing verdict (*no elementary witness*) is a statement about this test only, not a claim that N is open. Over the 232 orders with $q \leq 3000$, the test passes 131 (audited, Paley, or Turyn) and fails 101. The smallest failure is

$$q = 109, \quad N = 23980, \quad m = 5 \cdot 11 \cdot 109,$$

obstructed by 109 ($2 \cdot 109 - 1 = 217 = 7 \cdot 31$ is not a prime power). This $N = 23980$ is the smallest order of the family for which we exhibit neither an audited entry nor an elementary witness; we do not assert it is open, and indeed Williamson-type matrices of order 109, were they tabulated, would settle it at once (compare $q = 73$).

Table 1 lists all such candidate orders up to $q = 2917$. The previously discussed $q = 2689$ and $q = 2917$ appear at the end; both are obstructed (2689 and 2917 are primes with $2 \cdot 2689 - 1 = 5377 = 19 \cdot 283$ and $2 \cdot 2917 - 1 = 5833 = 19 \cdot 307$), and $q = 2917$ has the extreme form $m = 1459 \cdot 2917$, a product of two obstructing primes. They are not distinguished among the candidates except by size.

Table 1: Orders $N = 2q(q + 1)$, q a prime power $\equiv 1 \pmod{4}$, with no elementary witness (§A.3), $109 \leq q \leq 2917$. "obstr." lists the obstructing prime factors of the odd part $m = q(q + 1)/2$.

q	N	odd part m	obstr.
109	23980	$5 \cdot 11 \cdot 109$	109
173	60204	$3 \cdot 29 \cdot 173$	173
197	78012	$3^2 \cdot 11 \cdot 197$	197
277	154012	$139 \cdot 277$	277
389	303420	$3 \cdot 5 \cdot 13 \cdot 389$	389
397	316012	$199 \cdot 397$	397
409	335380	$5 \cdot 41 \cdot 409$	409
457	418612	$229 \cdot 457$	457
509	519180	$3 \cdot 5 \cdot 17 \cdot 509$	509
521	543924	$3^2 \cdot 29 \cdot 521$	521
593	704484	$3^3 \cdot 11 \cdot 593$	593
613	752764	$307 \cdot 613$	613
617	762612	$3 \cdot 103 \cdot 617$	103, 617

q	N	odd part m	obstr.
641	823044	$3 \cdot 107 \cdot 641$	107, 641
701	984204	$3^3 \cdot 13 \cdot 701$	701
733	1076044	$367 \cdot 733$	733
757	1147612	$379 \cdot 757$	757
797	1272012	$3 \cdot 7 \cdot 19 \cdot 797$	797
821	1349724	$3 \cdot 137 \cdot 821$	137, 821
829	1376140	$5 \cdot 83 \cdot 829$	83
853	1456924	$7 \cdot 61 \cdot 853$	853
937	1757812	$7 \cdot 67 \cdot 937$	67
953	1818324	$3^2 \cdot 53 \cdot 953$	953
1009	2038180	$5 \cdot 101 \cdot 1009$	101
1021	2086924	$7 \cdot 73 \cdot 1021$	73, 1021
1033	2136244	$11 \cdot 47 \cdot 1033$	1033
1097	2409012	$3^2 \cdot 61 \cdot 1097$	1097
1109	2461980	$3 \cdot 5 \cdot 37 \cdot 1109$	1109
1117	2497612	$13 \cdot 43 \cdot 1117$	1117
1129	2551540	$5 \cdot 113 \cdot 1129$	113, 1129
1153	2661124	$577 \cdot 1153$	1153
1181	2791884	$3 \cdot 197 \cdot 1181$	197, 1181
1213	2945164	$607 \cdot 1213$	1213
1217	2964612	$3 \cdot 7 \cdot 29 \cdot 1217$	1217
1249	3122500	$5^4 \cdot 1249$	1249
1289	3325620	$3 \cdot 5 \cdot 43 \cdot 1289$	1289
1321	3492724	$661 \cdot 1321$	1321
1361	3707364	$3 \cdot 227 \cdot 1361$	227, 1361
1373	3773004	$3 \cdot 229 \cdot 1373$	1373
1381	3817084	$691 \cdot 1381$	1381
1453	4225324	$727 \cdot 1453$	1453
1481	4389684	$3 \cdot 13 \cdot 19 \cdot 1481$	1481
1553	4826724	$3 \cdot 7 \cdot 37 \cdot 1553$	1553
1597	5104012	$17 \cdot 47 \cdot 1597$	1597
1601	5129604	$3^2 \cdot 89 \cdot 1601$	89, 1601
1613	5206764	$3 \cdot 269 \cdot 1613$	269, 1613
1637	5362812	$3^2 \cdot 7 \cdot 13 \cdot 1637$	1637
1709	5844780	$3^2 \cdot 5 \cdot 19 \cdot 1709$	1709
1721	5927124	$3 \cdot 7 \cdot 41 \cdot 1721$	1721
1733	6010044	$3 \cdot 17^2 \cdot 1733$	1733
1741	6065644	$13 \cdot 67 \cdot 1741$	67
1753	6149524	$877 \cdot 1753$	1753
1777	6319012	$7 \cdot 127 \cdot 1777$	127, 1777
1789	6404620	$5 \cdot 179 \cdot 1789$	179, 1789
1873	7020004	$937 \cdot 1873$	1873
1877	7050012	$3 \cdot 313 \cdot 1877$	1877
1889	7140420	$3^3 \cdot 5 \cdot 7 \cdot 1889$	1889
1933	7476844	$967 \cdot 1933$	1933
1997	7980012	$3^3 \cdot 37 \cdot 1997$	1997
2017	8140612	$1009 \cdot 2017$	2017
2053	8433724	$13 \cdot 79 \cdot 2053$	2053
2081	8665284	$3 \cdot 347 \cdot 2081$	347, 2081
2113	8933764	$7 \cdot 151 \cdot 2113$	151, 2113
2129	9069540	$3 \cdot 5 \cdot 71 \cdot 2129$	71, 2129
2161	9344164	$23 \cdot 47 \cdot 2161$	2161
2213	9799164	$3^3 \cdot 41 \cdot 2213$	2213
2269	10301260	$5 \cdot 227 \cdot 2269$	227, 2269
2273	10337604	$3 \cdot 379 \cdot 2273$	2273
2281	10410484	$7 \cdot 163 \cdot 2281$	163
2293	10520284	$31 \cdot 37 \cdot 2293$	2293
2297	10557012	$3 \cdot 383 \cdot 2297$	383, 2297

q	N	odd part m	obstr.
2309	10667580	$3 \cdot 5 \cdot 7 \cdot 11 \cdot 2309$	2309
2341	10965244	$1171 \cdot 2341$	2341
2357	11115612	$3^2 \cdot 131 \cdot 2357$	131, 2357
2377	11305012	$29 \cdot 41 \cdot 2377$	2377
2381	11343084	$3 \cdot 397 \cdot 2381$	397, 2381
2389	11419420	$5 \cdot 239 \cdot 2389$	239, 2389
2417	11688612	$3 \cdot 13 \cdot 31 \cdot 2417$	2417
2437	11882812	$23 \cdot 53 \cdot 2437$	2437
2441	11921844	$3 \cdot 11 \cdot 37 \cdot 2441$	2441
2473	12236404	$1237 \cdot 2473$	2473
2477	12276012	$3 \cdot 7 \cdot 59 \cdot 2477$	2477
2609	13618980	$3^2 \cdot 5 \cdot 29 \cdot 2609$	2609
2621	13744524	$3 \cdot 19 \cdot 23 \cdot 2621$	2621
2657	14124612	$3 \cdot 443 \cdot 2657$	443, 2657
2677	14338012	$13 \cdot 103 \cdot 2677$	103, 2677
2689	14466820	$5 \cdot 269 \cdot 2689$	269, 2689
2693	14509884	$3 \cdot 449 \cdot 2693$	449, 2693
2713	14726164	$23 \cdot 59 \cdot 2713$	2713
2749	15119500	$5^3 \cdot 11 \cdot 2749$	2749
2777	15429012	$3 \cdot 463 \cdot 2777$	463, 2777
2789	15562620	$3^2 \cdot 5 \cdot 31 \cdot 2789$	2789
2797	15652012	$1399 \cdot 2797$	2797
2801	15696804	$3 \cdot 467 \cdot 2801$	467, 2801
2833	16057444	$13 \cdot 109 \cdot 2833$	109, 2833
2837	16102812	$3 \cdot 11 \cdot 43 \cdot 2837$	2837
2857	16330612	$1429 \cdot 2857$	2857
2909	16930380	$3 \cdot 5 \cdot 97 \cdot 2909$	2909
2917	17023612	$1459 \cdot 2917$	2917

None of the orders in Table 1 is a known open Hadamard order; by (4.1) each odd part is composite, while the open orders below 10^4 are $4 \times \text{prime}$. Widening the ingredient set (computer-tabulated Williamson and T -matrices, or Goethals–Seidel cyclotomic difference sets) removes candidates from the table but can never produce one that meets the open list. The role of Theorem 1 is thus to give a single closed-form rule for the whole family, including the region these tables do not reach, rather than to settle any open existence question.

References

- [1] A. Farouk and Q.-W. Wang, An infinite family of Hadamard matrices constructed from Paley type matrices, *Filomat* 34 (2020), no. 3, 815–834.
- [2] A. Farouk and Q.-W. Wang, Construction of new Hadamard matrices using known Hadamard matrices, *Filomat* 36 (2022), no. 6, 2025–2042.
- [3] J. Hadamard, Résolution d’une question relative aux déterminants, *Bull. Sci. Math.* 17 (1893), 240–246.
- [4] R. E. A. C. Paley, On orthogonal matrices, *J. Math. Phys.* 12 (1933), 311–320.
- [5] U. Scarpis, Sui determinanti di valore massimo, *Rend. R. Ist. Lombardo Sci. Lett.* (2) 31 (1898), 1441–1446.
- [6] J. Williamson, Hadamard’s determinant theorem and the sum of four squares, *Duke Math. J.* 11 (1944), 65–81.

- [7] R. J. Turyn, Hadamard matrices, Baumert–Hall units, four-symbol sequences, pulse compression, and surface wave encodings, *J. Combin. Theory Ser. A* 16 (1974), 313–333.
- [8] J. Seberry and M. Yamada, Hadamard matrices, sequences, and block designs, in *Contemporary Design Theory* (J. H. Dinitz and D. R. Stinson, eds.), Wiley, 1992, pp. 431–560.
- [9] D. Z. Đoković, Generalization of Scarpis’s theorem on Hadamard matrices, arXiv:1601.00635, 2016. <https://arxiv.org/abs/1601.00635>.
- [10] H. Kharaghani and B. Tayfeh-Rezaie, A Hadamard matrix of order 428, *J. Combin. Des.* 13 (2005), 435–440. <https://doi.org/10.1002/jcd.20043>.
- [11] J. Seberry and M. Yamada, *Hadamard Matrices: Constructions using Number Theory and Linear Algebra*, Wiley, 2020. <https://doi.org/10.1002/9781119520252>.
- [12] M. Cati and D. V. Pasechnik, A database of constructions of Hadamard matrices, arXiv:2411.18897v2, 2025. <https://arxiv.org/abs/2411.18897>. <https://doi.org/10.48550/arXiv.2411.18897>.